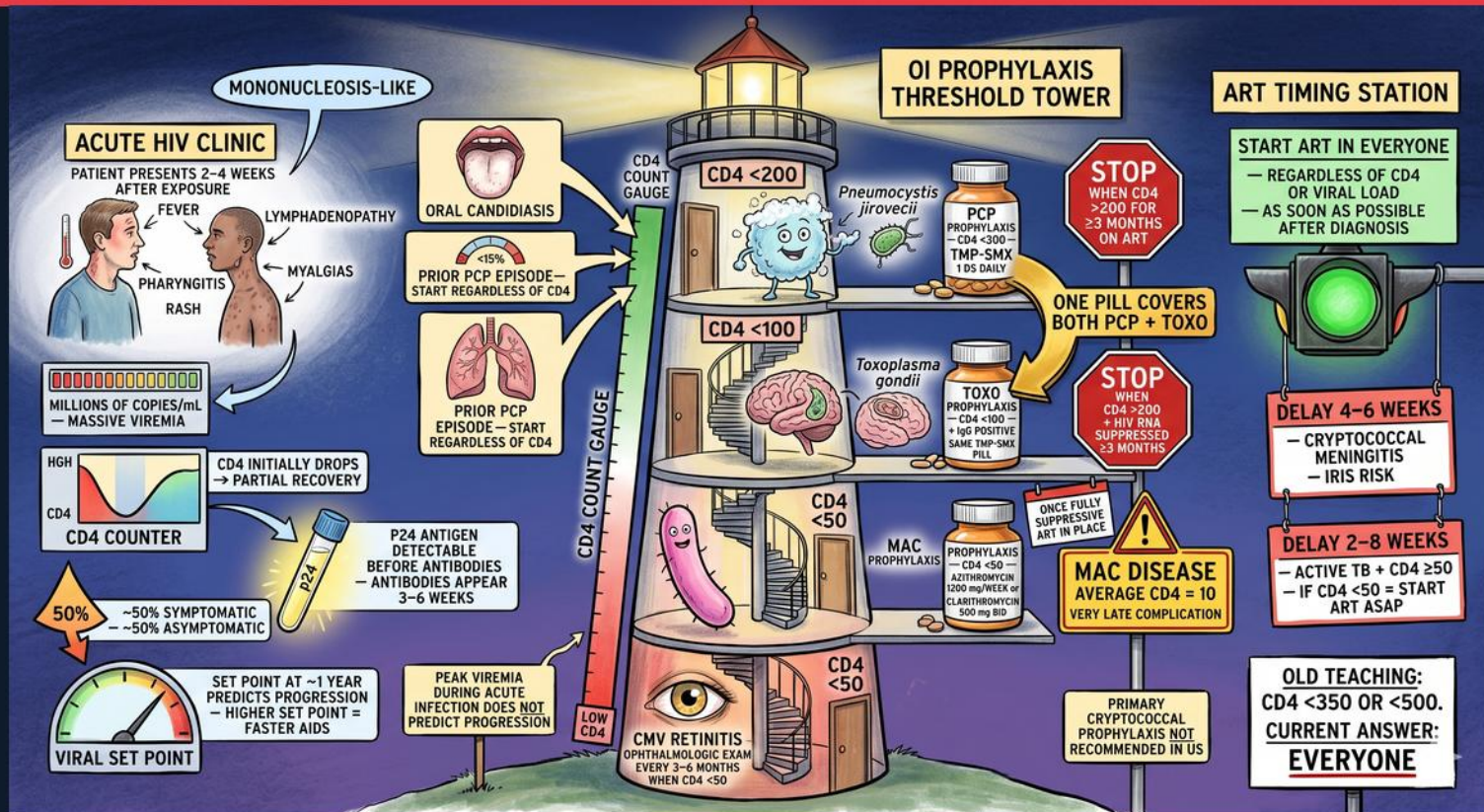
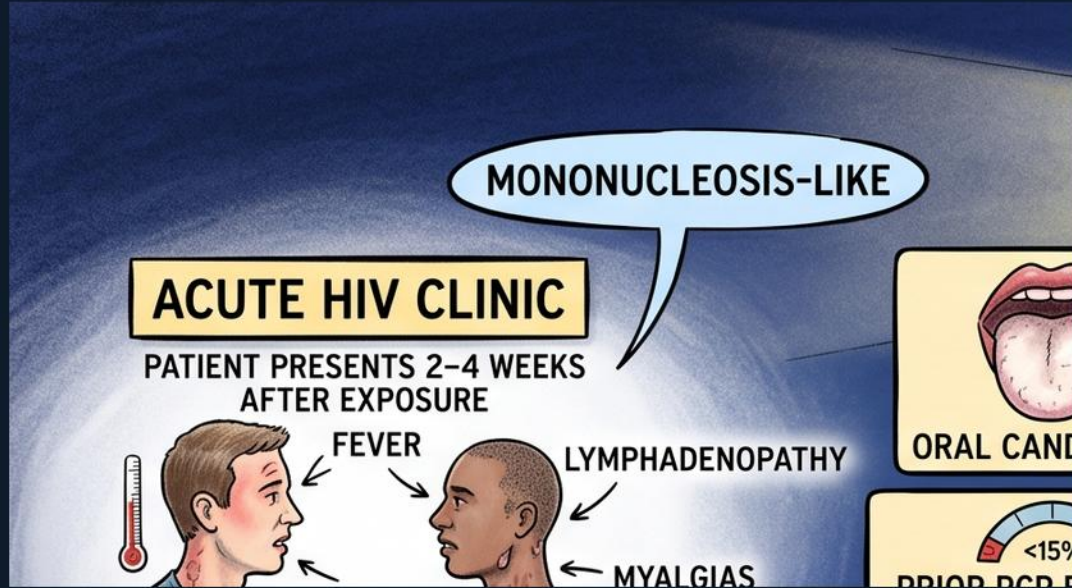


HIV — Acute Retroviral Syndrome, Diagnosis & Viral Set Point





Acute HIV clinic

WHAT YOU SEE

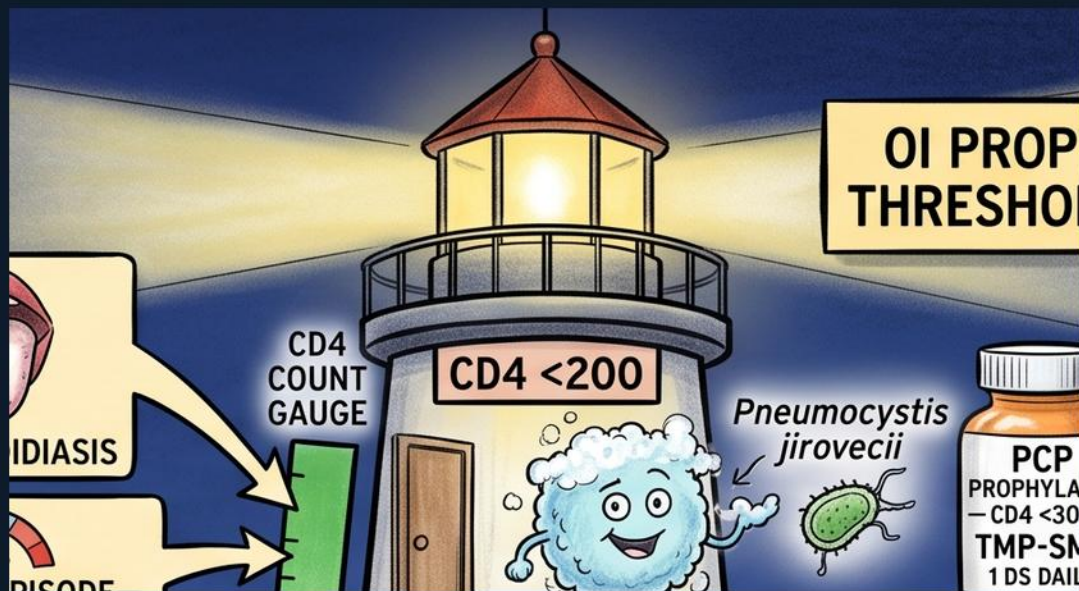
'Acute HIV Clinic' box, patient presents 2-4 weeks after exposure

WHAT IT ENCODES

Acute (primary) HIV / acute retroviral syndrome appears roughly 2–4 weeks after exposure as a self-limited mononucleosis-like illness, coinciding with peak viremia and seroconversion. It is one of the most underdiagnosed presentations because symptoms are nonspecific and the standard antibody-only test may still be negative. Have a low threshold to test for acute HIV in any compatible illness with risk exposure, because diagnosis here both prevents transmission (patient is maximally infectious) and allows early ART.

THE BOARD TRAP

Wrong answer: 'a negative 3rd-gen HIV antibody test 2 weeks after exposure rules out acute HIV.' Discriminator: this is the WINDOW PERIOD — antibodies haven't formed yet. The correct next step is a 4th-generation Ag/Ab combo assay or, if still suspicious, HIV-1 RNA (viral load) PCR, which turns positive earliest.



Mononucleosis-like

WHAT YOU SEE

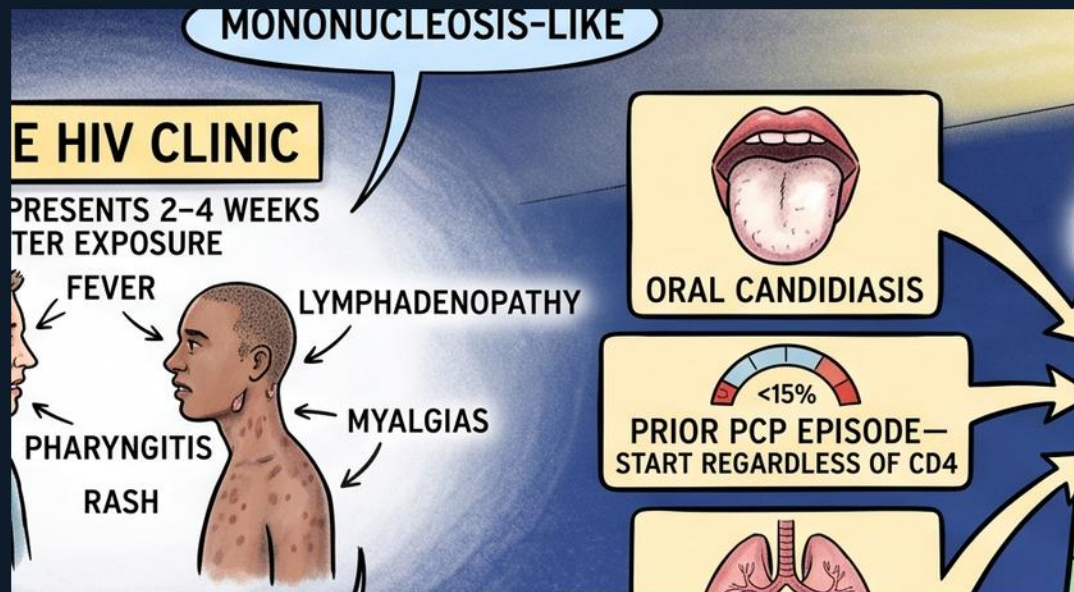
'Mononucleosis-like' banner

WHAT IT ENCODES

Acute HIV closely mimics infectious mononucleosis (EBV) and CMV: fever, sore throat, lymphadenopathy, fatigue. Features that push toward HIV over EBV mono include a maculopapular rash, painful mucocutaneous ulcers (oral/genital), and a heterophile (Monospot)-negative result in a person with risk factors. Always frame mono-like illness in a sexually active or high-risk patient as 'rule out acute HIV.'

THE BOARD TRAP

Wrong answer: 'heterophile-negative mononucleosis in a sexually active patient → it's just EBV-negative CMV mono, no HIV testing needed.' Discriminator: heterophile-negative mono-like illness with rash or mucocutaneous ulcers in an at-risk patient is acute HIV until proven otherwise — order a 4th-gen Ag/Ab test plus HIV RNA.



Fever / pharyngitis / rash

WHAT YOU SEE

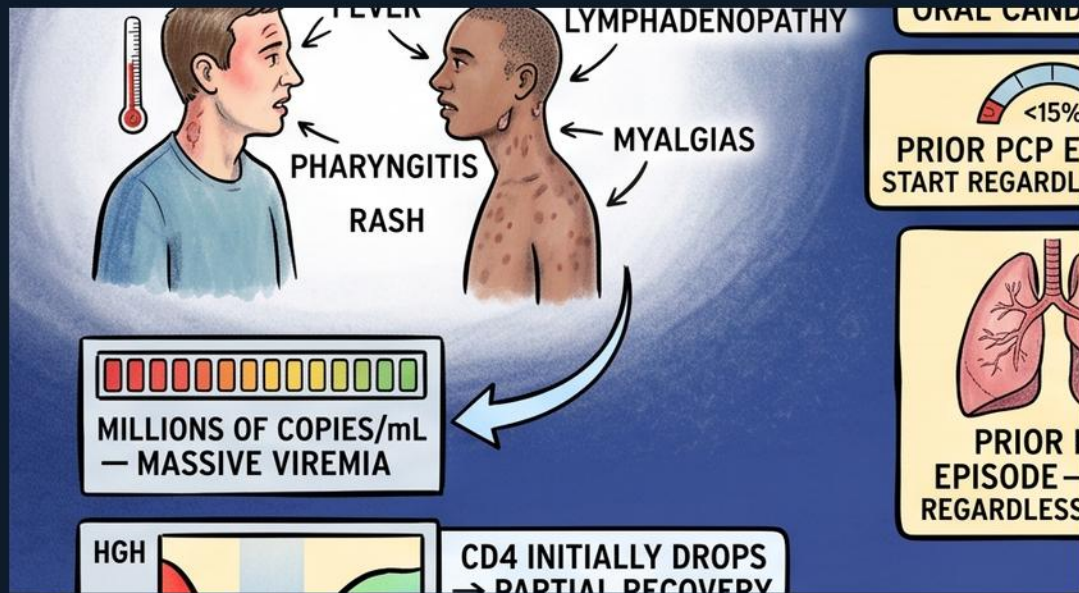
Patient with fever, lymphadenopathy, pharyngitis, myalgia, rash

WHAT IT ENCODES

The classic acute retroviral syndrome pentad: fever (most common, >90%), generalized lymphadenopathy, pharyngitis, a maculopapular/morbilliform rash (often trunk/face), and myalgia/arthritis, frequently with headache and painful mucocutaneous ulcers. Symptoms typically last 1–3 weeks then resolve spontaneously, which falsely reassures patient and clinician.

THE BOARD TRAP

Wrong answer: 'the rash of acute HIV is vesicular like varicella.' Discriminator: acute HIV rash is a non-pruritic maculopapular/morbilliform eruption on the trunk and face, classically with painful oral/genital ulcers — not vesicular. A vesicular rash points elsewhere (VZV/HSV).



Massive viremia

WHAT YOU SEE

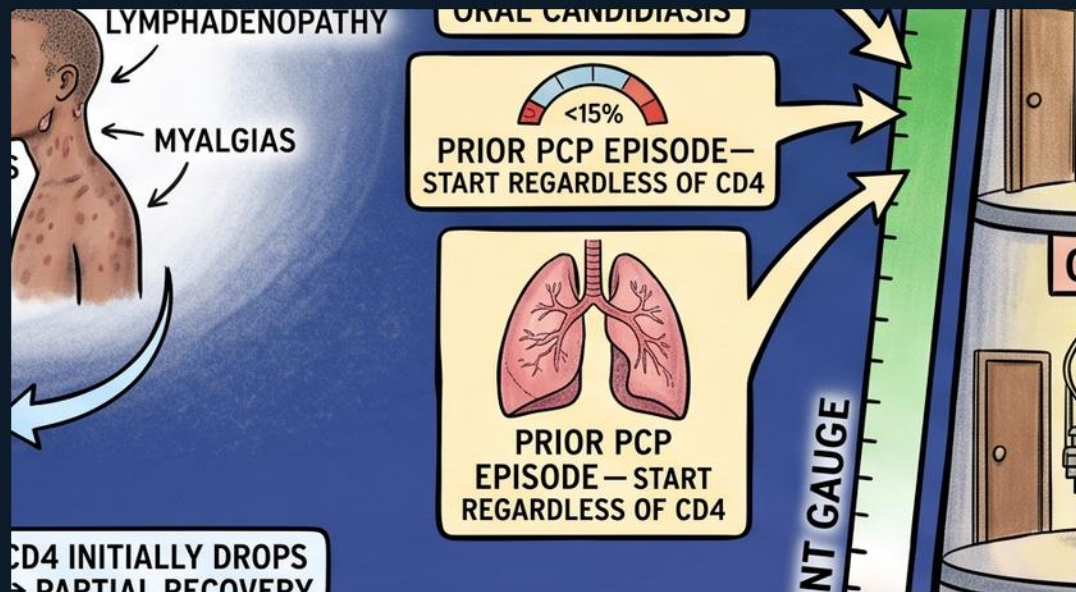
'Millions of copies/mL — massive viremia'

WHAT IT ENCODES

During acute infection the plasma HIV RNA is extraordinarily high — often millions of copies/mL — because there is no adaptive immune control yet. This is why HIV RNA is the earliest detectable marker (positive ~10 days post-exposure, before p24 antigen and well before antibodies) and why patients in acute infection are the most contagious and contribute disproportionately to onward transmission.

THE BOARD TRAP

Wrong answer: 'order a CD4 count to diagnose acute HIV.'
 Discriminator: CD4 is for staging and OI-prophylaxis decisions, NOT diagnosis. In the acute window the diagnostic test is HIV RNA (or 4th-gen Ag/Ab); CD4 can even be transiently low and is uninformative for whether acute HIV is present.



CD4 transient drop

WHAT YOU SEE

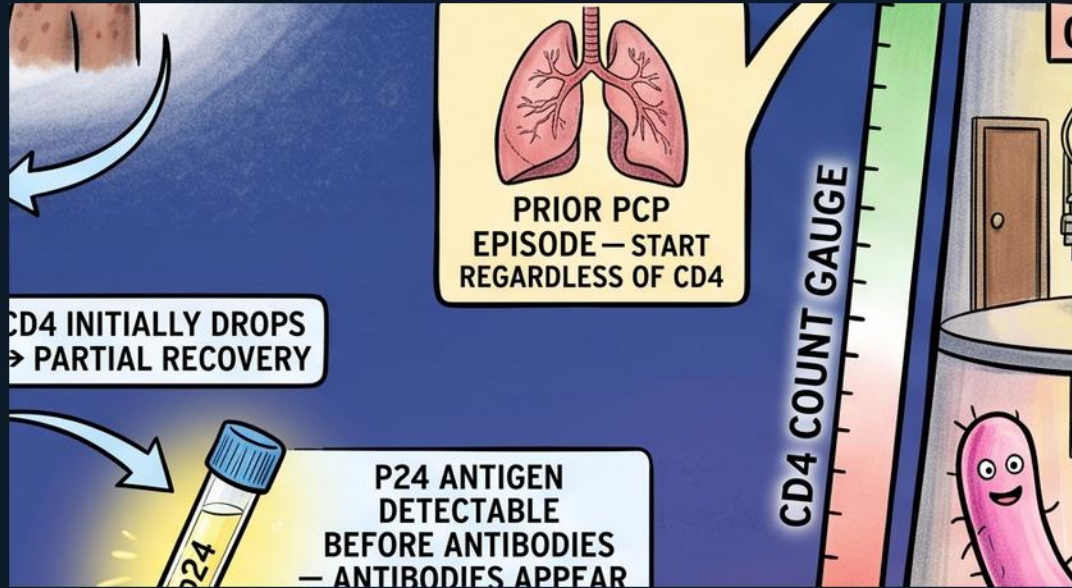
'CD4 initially drops → partial recovery'

WHAT IT ENCODES

In acute infection CD4 cells are killed rapidly and the count transiently falls (occasionally low enough to permit an early opportunistic infection), then **PARTIALLY** recovers once the immune response establishes the viral set point. Thereafter CD4 declines slowly over years (~50–100 cells/μL/year untreated) until immunodeficiency. The transient acute dip does not equal AIDS.

THE BOARD TRAP

Wrong answer: 'a low CD4 during acute HIV means the patient already has AIDS.' Discriminator: the acute CD4 dip is transient and recovers; AIDS staging is based on the sustained chronic decline (CD4 <200) or an AIDS-defining illness — not the acute-phase nadir.



p24 antigen window

WHAT YOU SEE

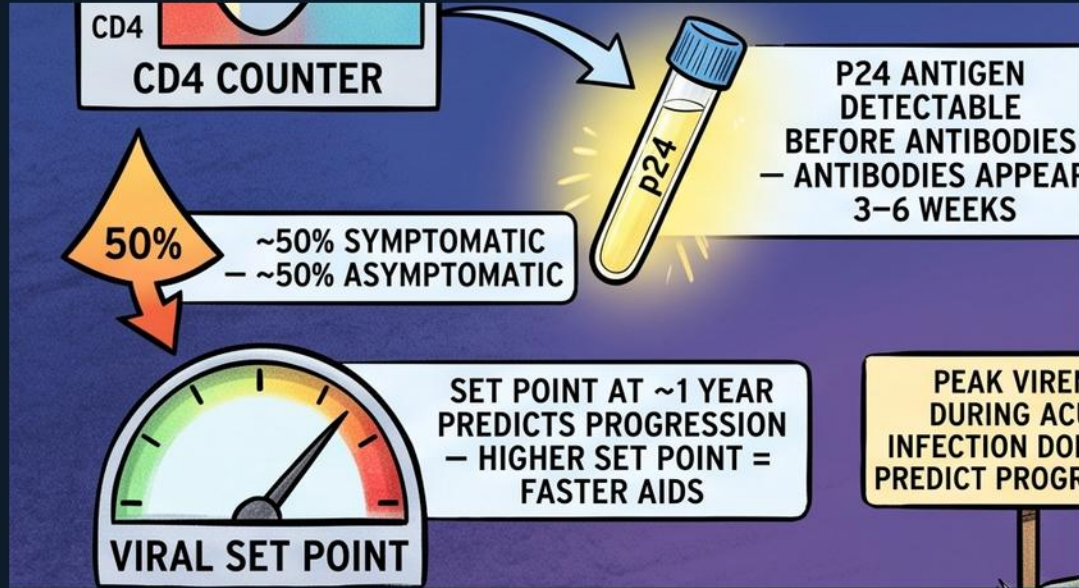
'p24 antigen detectable before antibodies; antibodies appear 3-6 weeks'

WHAT IT ENCODES

Marker timeline: HIV RNA detectable ~10 days → p24 antigen ~2–3 weeks → antibodies ~3–6 weeks (some up to 12). Fourth-generation combination assays detect BOTH p24 antigen and HIV-1/2 antibodies, shrinking the window to ~2–3 weeks. The CDC algorithm: 4th-gen Ag/Ab screen → if reactive, HIV-1/2 antibody differentiation immunoassay → if discordant/negative, HIV-1 RNA.

THE BOARD TRAP

Wrong answer: 'use a 3rd-generation antibody-only test to screen for early HIV.' Discriminator: antibody-only (3rd-gen) assays miss the window period. The standard is a 4th-gen Ag/Ab combo (catches p24 antigen weeks earlier); if acute infection is strongly suspected and the 4th-gen is negative, add HIV RNA.



Viral set point

WHAT YOU SEE

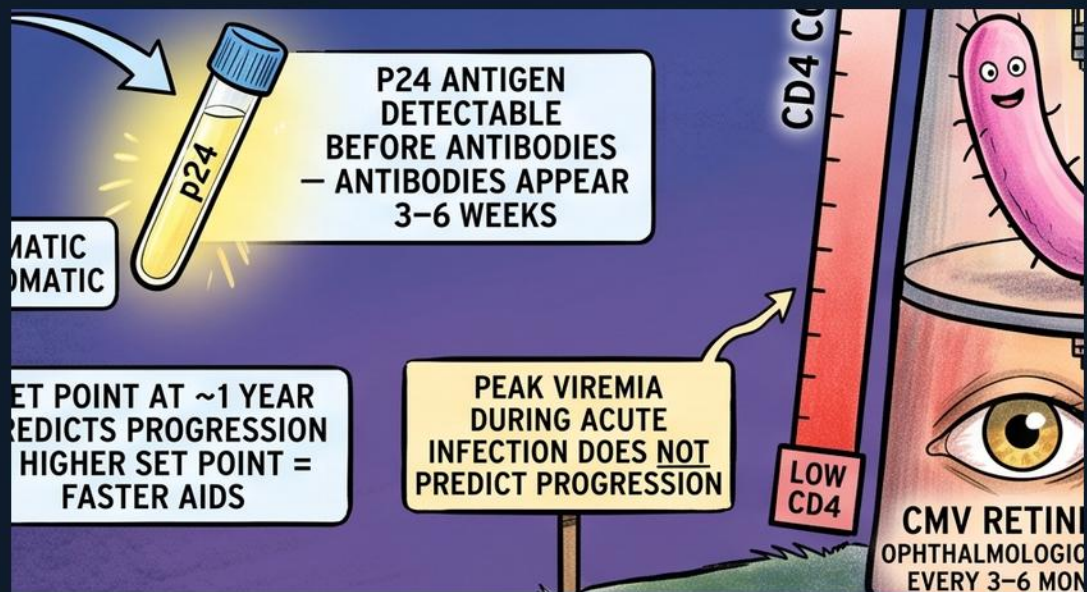
'Viral Set Point' tube and label

WHAT IT ENCODES

After the acute viremic spike, the cytotoxic T-cell/immune response brings viral load down to a stable plateau — the viral set point — reached roughly 6 months after infection. It reflects the equilibrium between viral replication and host immune control and remains relatively constant during the long asymptomatic/clinical-latency phase that follows.

THE BOARD TRAP

Wrong answer: 'the viral set point is the highest viral load seen during infection.' Discriminator: the set point is the stable PLATEAU reached after acute infection (~6 months), not the acute peak. The acute peak is transiently much higher and is brought down to the set point by the immune response.



Set point predicts progression

WHAT YOU SEE

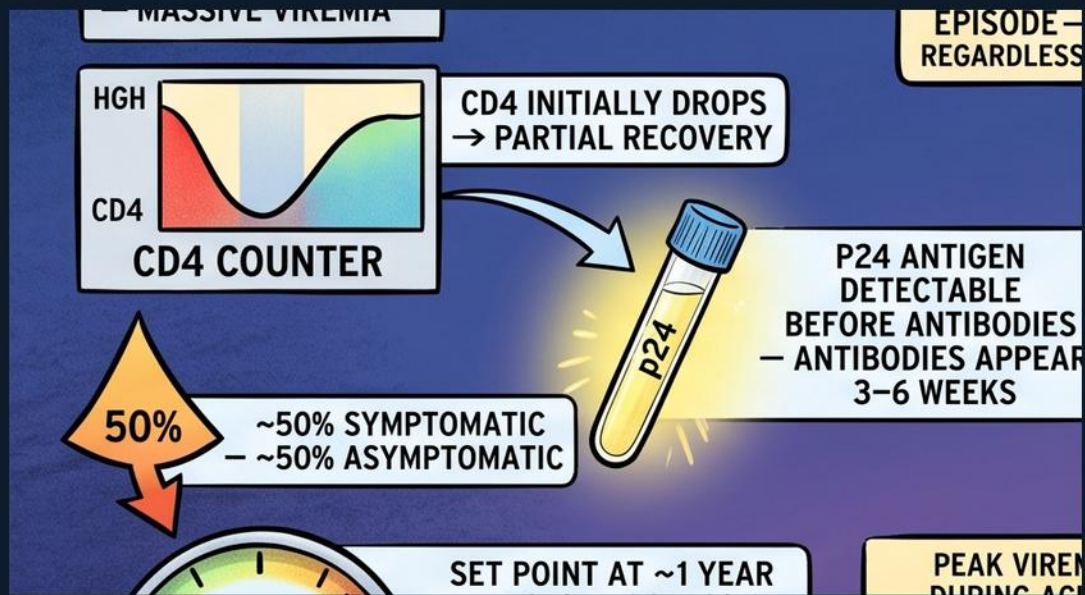
'Set point at ~1 year predicts progression; higher = faster AIDS'

WHAT IT ENCODES

The viral set point (around the 1-year mark) is the single best untreated predictor of disease course: a HIGHER set point predicts faster CD4 decline and faster progression to AIDS, while a low set point predicts slow progression (long-term non-progressors). This is why establishing durable virologic suppression with ART changes the trajectory entirely.

THE BOARD TRAP

Wrong answer: 'CD4 count, not viral load, predicts the RATE of progression to AIDS.' Discriminator: CD4 measures CURRENT immune status, but the viral SET POINT predicts the RATE/velocity of future decline. Both matter, but for prognostic speed of progression the set point is the classic answer.



~50% symptomatic

WHAT YOU SEE

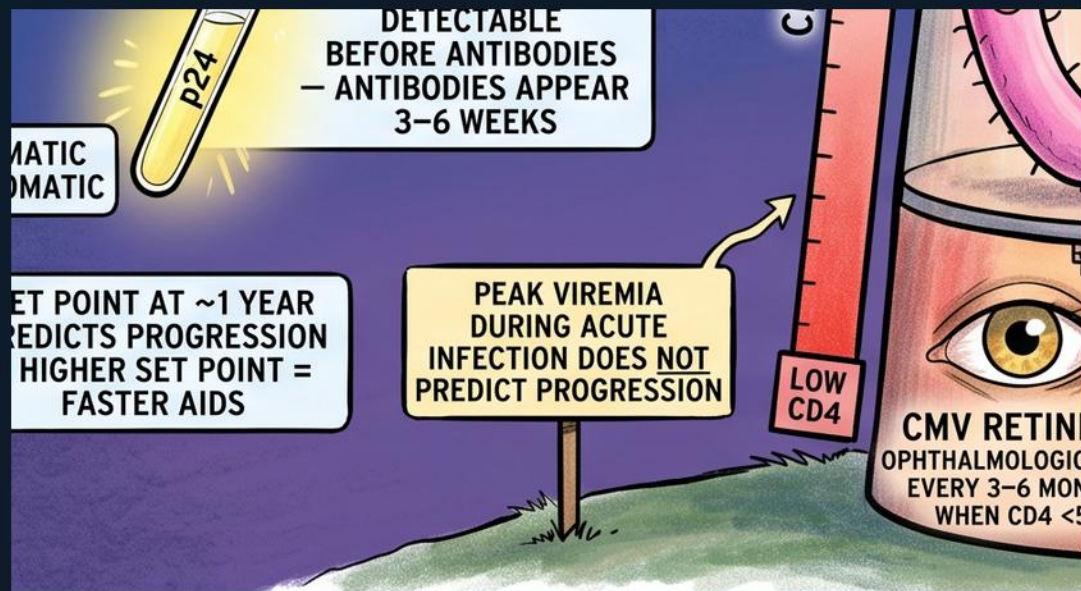
'50% — ~50% symptomatic / ~50% asymptomatic'

WHAT IT ENCODES

Only roughly half of people with acute HIV have a clinically recognized illness; the rest are asymptomatic or so mild they don't seek care. Symptomatic acute infection (especially prolonged/severe) is associated with a higher set point and faster progression, but absence of symptoms in no way excludes infection — testing, not symptoms, drives diagnosis.

THE BOARD TRAP

Wrong answer: 'the patient felt completely well after exposure, so acute HIV is excluded.' Discriminator: about 50% of acute infections are asymptomatic; feeling well does NOT rule out HIV. Risk-based testing (4th-gen Ag/Ab ± RNA) is required regardless of symptoms.



Peak viremia infectivity

WHAT YOU SEE

'Peak viremia during acute infection does NOT predict progression'

WHAT IT ENCODES

The very high PEAK viremia of acute infection drives intense transmissibility (a major reason undiagnosed acute infection fuels spread), but it does NOT predict long-term outcome. It is the post-acute SET POINT — not the transient acute peak — that correlates with the rate of progression to AIDS.

THE BOARD TRAP

Wrong answer: 'a sky-high viral load during acute infection means rapid progression to AIDS.' Discriminator: the acute PEAK reflects infectivity, not prognosis; the SET POINT (the stabilized plateau) is what predicts progression. Don't conflate the transient peak with the durable set point.

Summary — every symbol

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2. Mononucleosis-like

Acute HIV closely mimics infectious mononucleosis (EBV) and CMV: fever, sore throat, lymphadenopathy, fatigue

3. Fever / pharyngitis / rash

The classic acute retroviral syndrome pentad: fever (most common, >90%), generalized lymphadenopathy, pharyngitis, a maculopapular/morbilliform rash (often trunk/face), and myalgia/arthralgia, frequently with headache and painful mucocutaneous ulcers

4. Massive viremia

During acute infection the plasma HIV RNA is extraordinarily high — often millions of copies/mL — because there is no adaptive immune control yet

5. CD4 transient drop

In acute infection CD4 cells are killed rapidly and the count transiently falls (occasionally low enough to permit an early opportunistic infection), then PARTIALLY recovers once the immune response establishes the viral set point

6. p24 antigen window

Marker timeline: HIV RNA detectable ~10 days → p24 antigen ~2–3 weeks → antibodies ~3–6 weeks (some up to 12)

7. Viral set point

After the acute viremic spike, the cytotoxic T-cell/immune response brings viral load down to a stable plateau — the viral set point — reached roughly 6 months after infection

8. Set point predicts progression

The viral set point (around the 1-year mark) is the single best untreated predictor of disease course: a HIGHER set point predicts faster CD4 decline and faster progression to AIDS, while a low set point predicts slow progression (long-term non-progressors)

9. ~50% symptomatic

Only roughly half of people with acute HIV have a clinically recognized illness; the rest are asymptomatic or so mild they don't seek care

10. Peak viremia infectivity

The very high PEAK viremia of acute infection drives intense transmissibility (a major reason undiagnosed acute infection fuels spread), but it does NOT predict long-term outcome